

GLT-STEER: Final-Marker Directions as Activation-Space Editors

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Abstract

This short draft studies whether mean hidden-state directions learned from sentence-pair differences can be injected back into a generative transformer as activation-space editors. In GPT-2, directions derived from controlled final-marker pairs steer surface markers such as ?, !, and ... under matched controls. The claim is intentionally bounded: current GLT-STEER supports final-marker form steering, not general semantic rewriting, and a norm-matched ? token direction can reproduce the learned question-delta rate.

1. Question and Scope

For a source sentence x and transformed sentence y , GLT computes a hidden-state displacement $\delta = h(y) - h(x)$. Earlier GLT tracks test whether these deltas classify transformation type offline. GLT-STEER asks a stronger intervention question: if a transformation δ is injected during generation, does the model produce the corresponding transformation marker?

The current defensible claim is that activation-space directions can steer final-position surface markers in GPT-style residual streams. The cleanest evidence is in GPT-2, with marker-dependent transfer to DistilGPT-2. The current controls do not establish that a learned sentence-pair δ is unique or necessary for this effect. This is not a claim of robust semantic rewriting.

2. Method

- Build synthetic source/target pairs for a transformation class.
- Extract hidden states at selected GPT-style residual blocks.
- Compute a centroid transformation direction from training pairs.
- During generation, add the vector to the residual stream.
- Compare target steering against no-steering, wrong-marker, random-norm, negative-vector, and token-direction controls.

```
delta = h(target sentence) - h(source sentence)
```

```
y_pred generation: residual[layer] += gain * centroid_delta
```

3. Main Evidence

3.1 Question steering and prompt-only control

condition	question mark rate
historical tuned final-marker direction	0.9350
random-norm control	0.0000
wrong-class control	0.0000
negative-target control	0.0000

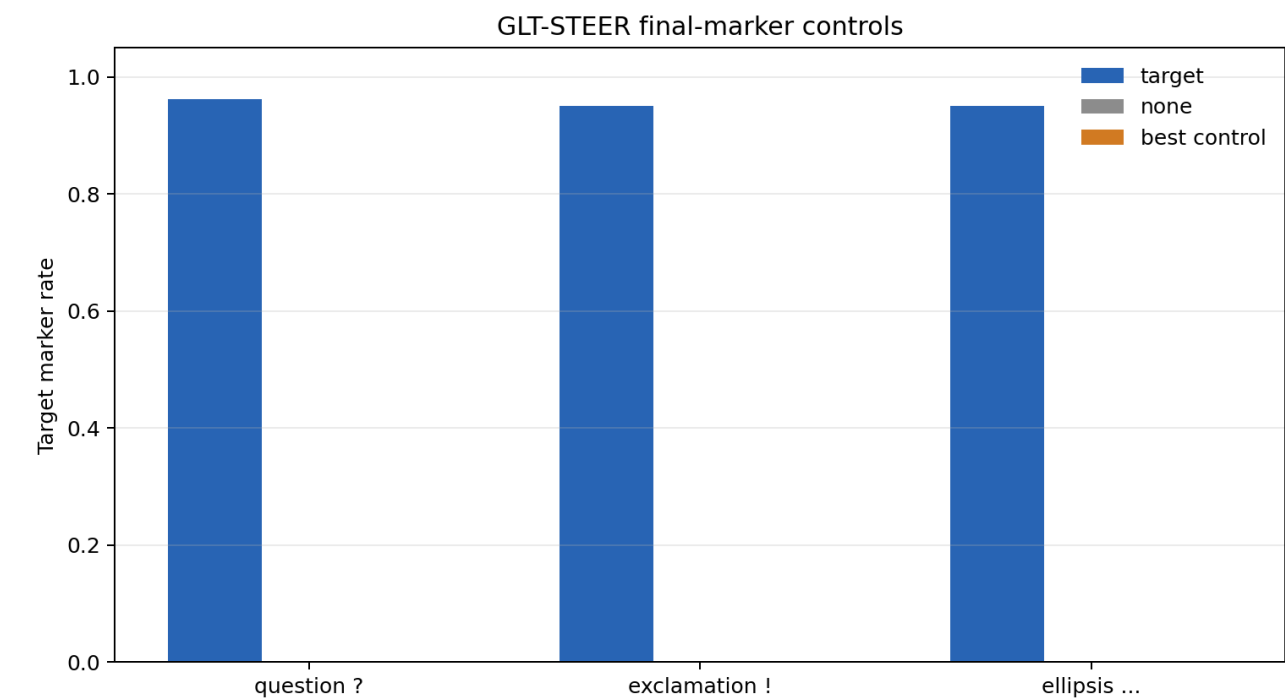
Copy-like prompts without steering produce 0.0000 question marks across 960 no-steering rows from GPT-2 and DistilGPT-2. This rules out the simplest prompt-only explanation for the question-marker result.

3.2 Content preservation

source set	prompt family	target question-and-preserved
in-template	copy-like prompts	0.9625-0.9750
simple out-of-template	copy-like prompts	0.8250-0.9000

Hard out-of-template sources reduce preservation. GPT-2 still shows question-form steering; DistilGPT-2 is weaker and should be reported as marker-form transfer only.

Figure 1. Final-marker controls



Exclamation and ellipsis controls show that the question result is part of a broader final-marker phenomenon, not only a question-specific artifact.

4. Mechanistic Audits

4.1 Logit-level final-marker audit

target	control	marker rate (N, 95% CI)	median best rank
?	none	0.0000 (N=96, [0.0000, 0.0385])	28.0
?	target	0.8542 (N=96, [0.7700, 0.9111])	1.0
!	none	0.0000 (N=96, [0.0000, 0.0385])	20.0
!	target	0.9063 (N=96, [0.8313, 0.9499])	1.0
...	none	0.0000 (N=96, [0.0000, 0.0385])	11.75
...	target	0.8750 (N=96, [0.7941, 0.9270])	1.0

No-steering marker rates are zero, while target steering moves the intended marker token to rank 1 in most sequences. This is the strongest current evidence that the intervention changes the generation distribution itself.

4.2 Position-of-intervention audit

condition	position mode	question rate	question+preserved
none	none	0.0000	0.0000
target_last_each_step	last_each_step	0.9604	0.7917
target_prompt_all_once	prompt_all	0.8625	0.7896
target_prompt_first	prompt_first	0.0000	0.0000
target_prompt_middle	prompt_middle	0.0000	0.0000
target_prompt_last_once	prompt_last	0.0000	0.0000

Single prompt-position edits do not work. Distributed prompt-state editing or repeated last-token-at-each-step editing does work. This argues against a trivial one-token prompt perturbation story.

5. Fixed-Parameter Confirmation

model	target	target marker rate	max control marker	target marker+preserved	max control marker+preserved
distilgpt2	ellipsis	0.8333 (N=144, CI [0.764, 0.885])	0.0000	0.1597 (N=144, CI [0.109, 0.228])	0.0000
distilgpt2	exclamation	0.8958 (N=144, CI [0.835, 0.936])	0.0139	0.3472 (N=144, CI [0.274, 0.428])	0.0000
distilgpt2	question	0.6319 (N=144, CI [0.551, 0.706])	0.0069	0.0556 (N=144, CI [0.028, 0.106])	0.0000
gpt2	ellipsis	0.8438 (N=288, CI [0.797, 0.881])	0.0000	0.3854 (N=288, CI [0.331, 0.443])	0.0000
gpt2	exclamation	0.6875 (N=288, CI [0.632, 0.738])	0.0000	0.3958 (N=288, CI [0.341, 0.453])	0.0000
gpt2	question	0.6562 (N=288, CI [0.600, 0.709])	0.0000	0.2396 (N=288, CI [0.194, 0.292])	0.0000

This run uses fresh hard-heldout sources and fixed settings: GPT-2 layers 2,3 at gain 0.75; DistilGPT-2 layer 2 at gain 1.0. No layer/gain search is performed inside the run. Target marker rates remain separated from controls across all three final markers and both tested models.

A source-cluster bootstrap treats the 48 sentence sources as the independent units rather than treating repeated prompt and layer rows as independent. The GPT-2 marker-rate estimates and percentile 95% cluster intervals are ?=0.6562 [0.5556, 0.7501], !=0.6875 [0.5764, 0.7882], and ...=0.8438 [0.7604, 0.9132]. These intervals are descriptive and do not remove the synthetic-template limitation.

5.1 Token-direction specificity control

control	question rate	marker+preserved
target delta	0.3750	0.0500
shuffled-pair delta	0.0500	0.0250
raw ? embedding	0.0000	0.0000
norm-matched ? embedding	0.3750	0.0750
other punctuation / random controls	0.0000	0.0000

The learned target delta is stronger than shuffled-pair and unrelated controls, but a ? embedding rescaled to the learned delta norm reproduces the same question-marker rate. The current evidence therefore supports activation-space final-marker induction, not a uniquely linguistic question-transformation vector.

6. Boundary Results

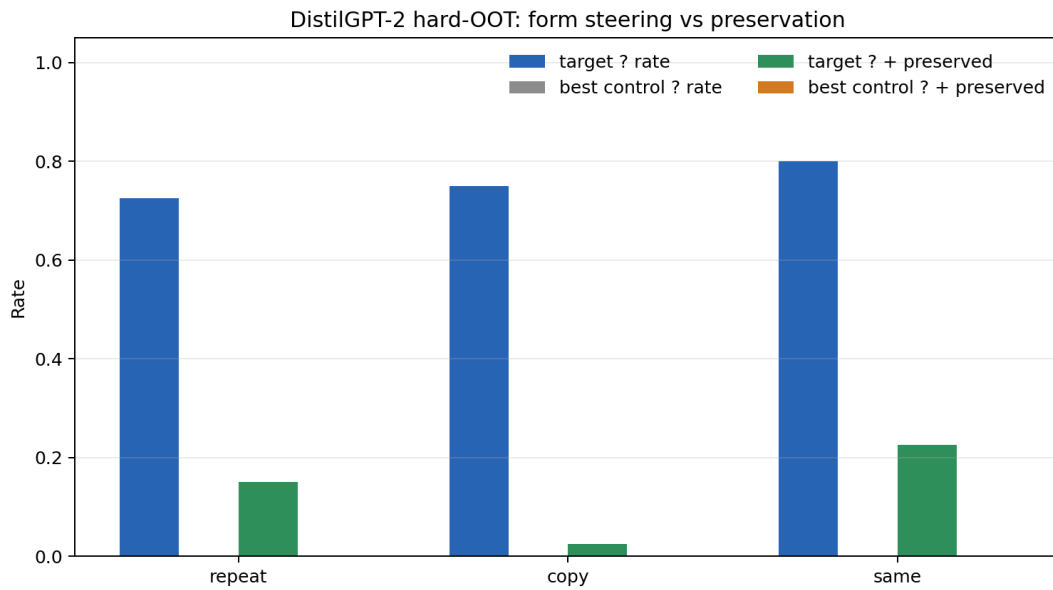
6.1 Negation

Negation is weak under the same copy-prompt recipe. A GPT-2 layer sweep over layers 0-11 does not find a clean intervention site. The best target-and-preserved row reaches only 0.1729, while matched controls remain nontrivial, with maxima up to 0.1229.

layer	question mean pairwise cosine	negation mean pairwise cosine
2	0.9693	0.5621
3	0.9365	0.5665

6.2 DistilGPT-2

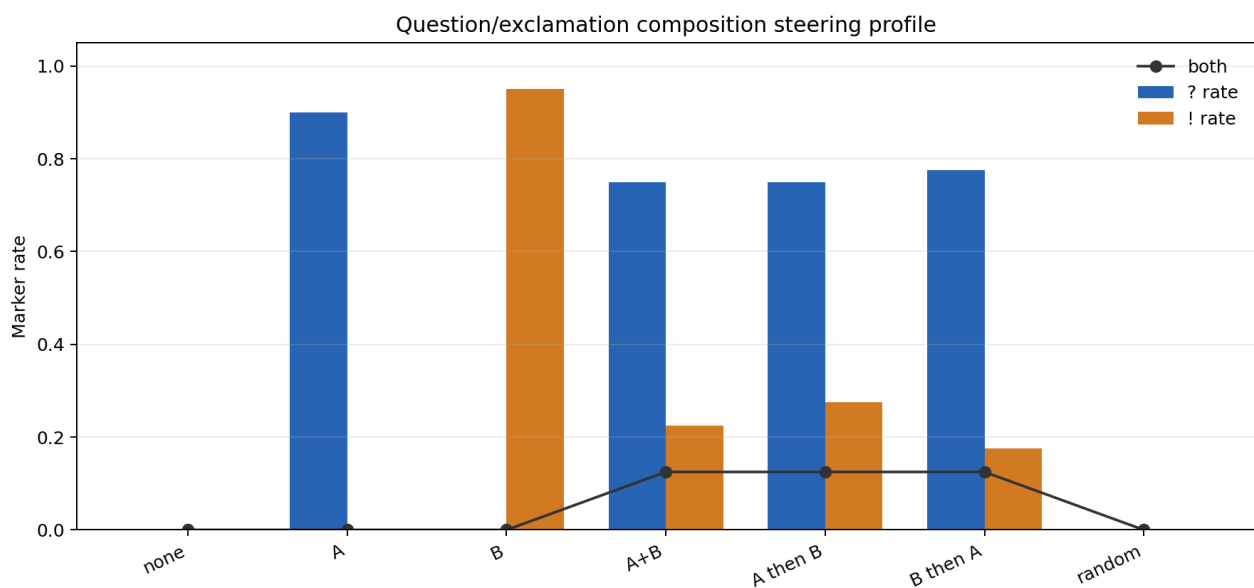
Figure 2. DistilGPT-2 hard-OOT boundary



DistilGPT-2 preserves marker-form steering but has weak strict content preservation on hard out-of-template sources.

6.3 Marker composition

Figure 3. Marker composition



Combined question/exclamation vectors produce mixed marker profiles and lower preservation. The correct interpretation is competition/saturation, not noncommutative algebra.

7. Related Work Positioning

GLT-STEER is closest to activation-engineering and representation-steering work. Activation Addition computes activation differences from contrastive prompts and injects them at inference time. Contrastive Activation Addition averages residual-stream differences over positive and negative behavioral examples. Representation Engineering frames a broader family of methods for monitoring and manipulating model internals.

The contribution here is not that activation addition exists. The narrower contribution is diagnostic: GLT-STEER derives vectors from controlled linguistic transformation pairs and asks which linguistic transformations behave as reusable intervention vectors. The current answer is bounded: final-position markers are steerable; sentence-internal transformations are not solved by this recipe.

8. Limitations

- GLT-STEER does not solve semantic rewriting.
- Negation and modality are not cleanly steered by the current method.
- Final-marker steering does not prove general linguistic transformation editing.
- The learned question delta is not unique relative to a norm-matched ? token direction.
- Marker composition does not prove noncommutative algebra.
- The result does not establish a Lie algebra in transformer activations.
- Prompt wording remains a real boundary condition.

9. Reproducibility

All reported results are archived in results/experiments/. No new model inference is required to inspect the tables in this draft. The datasets are synthetic controlled sentence-pair templates generated by repository scripts.

artifact	path
draft source	paper/articles/glt-steer-activation-editors/submission_draft.md
PDF builder	scripts/build_glt_steer_submission_pdf.py
generated PDF	reports/2026-08-25_glt_steer_submission_draft.pdf
confirmatory result	results/experiments/glt_steer_confirmatory_fixed_params_20260825_results/
CI audit	results/experiments/glt_steer_headline_ci_20260825_results/
source-cluster bootstrap	results/experiments/glt_steer_source_cluster_bootstrap_20261009_results/
token-direction controls	results/experiments/glt_steer_token_direction_controls_20261009_results/

References

- Turner et al. 2023/2024. Steering Language Models With Activation Engineering / Activation Addition. arXiv:2308.10248.
- Panickssery et al. 2024. Steering Llama 2 via Contrastive Activation Addition. arXiv:2312.06681 / ACL 2024.
- Zou et al. 2023. Representation Engineering: A Top-Down Approach to AI Transparency. arXiv:2310.01405.
- Frenor and Alvarez 2026. RISE: geometric rotations for semantic-syntactic transformations.
- Ilharco et al. 2023. Editing Models with Task Arithmetic.
- Todd et al. 2024. Function Vectors in Large Language Models.